A PROOF OF TANG AND ZHANG'S ORDER -2 SCHOENBERG CONJECTURE, WITH THE EQUALITY CASE

Abstract. Let $P(z) = z\prod_{j=1}^{n}(z - z_j)$ with $z_j \in \mathbb{C} \setminus \{0\}$, and let $w_1, \dots, w_n$ be the zeros of P' counted with multiplicity. Conjecture C1 of Tang and Zhang asserts that $\sum_k |w_k|^{-2} \le 3\sum_j |z_j|^{-2} + \left|\sum_j z_j^{-1}\right|^2$, and that equality holds whenever the z_j are collinear with the origin; they prove it for $n = 2$. We prove the inequality for every n, and show that equality holds *if and only if* $z_1, \dots, z_n$ lie on one line through the origin. Writing $U = \operatorname{diag}(z_1^{-1}, \dots, z_n^{-1})$ and J for the all-ones matrix, the proof conjugates the authors' matrix model $B = U(I + J)$, whose spectrum is $\{w_k^{-1}\}$, by the explicit square root $Q = (I+J)^{1/2} = I + \frac{\sqrt{n+1}-1}{n}J$: the conjectured right-hand side is exactly $\|QUQ\|_F^2$, so Schur's inequality with its equality case gives both halves at once, equality occurring precisely when QUQ is normal, which is precisely collinearity.

1. The conjecture

For a monic polynomial with a simple zero at the origin,

$$P(z) = z\prod_{j=1}^{n}(z - z_j), \qquad z_1, \dots, z_n \in \mathbb{C} \setminus \{0\} \quad \text{(repetitions allowed)}, \tag{1}$$

the derivative P' has degree exactly n and no zero at the origin; write $w_1, \dots, w_n$ for its zeros, *as a multiset*. Quanyu Tang and Teng Zhang state the following as Conjecture C1 of [4], in its section "Schoenberg type inequality of order -2".

Conjecture C1 (Tang–Zhang). Let $P(z) = z\prod_{j=1}^{n}(z - z_j)$, $z_j \in \mathbb{C} \setminus \{0\}$, and let $w_1, \dots, w_n$ denote the zeros of $P'(z)$. Then

$$\sum_{k=1}^{n} \frac{1}{|w_k|^2} \;\le\; 3\sum_{j=1}^{n} \frac{1}{|z_j|^2} + \left|\sum_{j=1}^{n} \frac{1}{z_j}\right|^2. \tag{2}$$

Moreover, equality in (2) holds whenever $z_1, \dots, z_n$ all lie on the same line passing through the origin.

Tang and Zhang prove (2) for $n = 2$.

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2. Notation and the result

Throughout, $n \geq 1$ and $z_1, \ldots, z_n$ are as in (1). Put

$$u_j = \frac{1}{z_j}, \quad U = \operatorname{diag}(u_1, \ldots, u_n), \quad S = \sum_{j=1}^{n} u_j, \quad D = \operatorname{diag}(z_1, \ldots, z_n), \tag{3}$$

$$J = \mathbf{1}\mathbf{1}^T \text{ (all ones)}, \quad M = I + J, \quad c = \frac{\sqrt{n+1} - 1}{n}, \quad Q = I + cJ, \tag{4}$$

$$K = QUQ, \quad \text{that is} \quad K_{jk} = \delta_{jk} u_j + c\,(u_j + u_k) + c^2 S. \tag{5}$$

Say that $z_1, \ldots, z_n$ are *collinear with the origin* if they all lie on a single line through 0, i.e. if $z_j = \rho r_j$ for some $\rho \in \mathbb{C} \setminus \{0\}$ and some real $r_j \neq 0$; equivalently, if $z_j \overline{z_k} \in \mathbb{R}$ for all j, k. For $n = 1$ the condition is vacuous.

Theorem 1. *For every $n \geq 1$ and all $z_1, \ldots, z_n \in \mathbb{C} \setminus \{0\}$, repetitions allowed, the inequality* (2) *holds. Equality holds in* (2) *if and only if $z_1, \ldots, z_n$ are collinear with the origin.*

Only the inequality and the “whenever” half of the equality clause are stated in [4]; the “only if” half does not appear there. We use Schur's inequality in the form in which it carries its own equality case.

Lemma 2 (Schur)**.** *For every $X \in \mathbb{C}^{n \times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ listed with multiplicity, $\sum_k |\lambda_k|^2 \leq \|X\|_F^2$, with equality if and only if X is normal.*

The inequality is [2]; for the equality case see [1, §2.5]. It is quoted in this form, with the equality case, in [4].

3. Proof of Theorem 1

Step 1: an explicit square root of $M = I + J$. Since $J^2 = nJ$,

$$Q^2 = (I + cJ)^2 = I + (2c + nc^2)J = I + J = M,$$

because $nc^2 + 2c - 1 = 0$ by the choice of c in (4). The matrix Q is real symmetric with positive entries, and it is invertible with $Q^{-1} = I + c'J$, $c' = (\frac{1}{\sqrt{n+1}} - 1)/n$, since $c + c' + ncc' = 0$. Note $M^2 = I + (n+2)J \neq M$: the sandwich matrix of the order -2 problem is not idempotent, which is why a square root is taken here rather than a projection inserted.

Step 2: the spectrum of K, as a multiset. Let $A = (I - \frac{1}{n+1}J)D$. Since JD is the rank-one matrix $\mathbf{1}z^T$, the Schur determinant formula gives

$$\det(wI - A) = \prod_{j=1}^{n} (w - z_j) \cdot \Big(1 + \tfrac{1}{n+1} \sum_{j=1}^{n} \tfrac{z_j}{w - z_j}\Big).$$

Substituting $\frac{z_j}{w - z_j} = -1 + \frac{w}{w - z_j}$ turns the bracket into $\frac{1}{n+1}\big(1 + w \sum_j \frac{1}{w - z_j}\big)$, and since $P'(w) = \prod_j (w - z_j) \cdot \big(1 + w \sum_j \frac{1}{w - z_j}\big)$ we get the identity

$$\det(wI - A) = \frac{P'(w)}{n+1}, \tag{6}$$

whose right-hand side is monic of degree n. Hence $\operatorname{spec}(A) = \{w_1, \dots, w_n\}$ *as a multiset*, and $\prod_k w_k = e_n(z)/(n+1) \neq 0$, so no w_k vanishes and A is invertible. Because $(I - \frac{1}{n+1}J)(I + J) = I$ exactly, we may read off the inverse:

$$B := A^{-1} = D^{-1}(I + J) = UM, \quad \operatorname{spec}(B) = \left\{\tfrac{1}{w_1}, \dots, \tfrac{1}{w_n}\right\} \text{ with mult.} \tag{7}$$

Finally $K = QUQ = Q(UM)Q^{-1} = QBQ^{-1}$ is similar to B, so $\operatorname{spec}(K) = \{w_k^{-1}\}$ with multiplicity, and

$$\sum_{k=1}^{n} \frac{1}{|w_k|^2} = \sum_{k=1}^{n} \big|\lambda_k(K)\big|^2. \tag{8}$$

The multiset reading in (6) is what (2) requires, since it sums over the n zeros of P' with multiplicity. At $z = (1,1,1)$ one has $P'(z) = 4z^3 - 9z^2 + 6z - 1 = (z-1)^2(4z-1)$, so $w = 1$ is a double zero and (2) closes as $1 + 1 + 16 = 18 = 9 + 9$.

Step 3: the conjectured right-hand side is $\|K\|_F^2$. Since Q is real symmetric, $K^* = Q\overline{U}Q$, so using $Q^2 = M$,

$$\begin{aligned}\|K\|_F^2 &= \operatorname{tr}(KK^*) = \operatorname{tr}(QUQ \cdot Q\overline{U}Q) = \operatorname{tr}(Q^2 UM\overline{U}) \\ &= \operatorname{tr}(MUM\overline{U}) = \sum_{j,m} M_{jm}^2\, u_m \overline{u_j}.\end{aligned}$$

Now $M_{jm}^2 = (1 + \delta_{jm})^2 = 1 + 3\delta_{jm}$ and $\sum_{j,m} u_m \overline{u_j} = |S|^2$, whence

$$\|K\|_F^2 = 3\sum_{j=1}^{n} |u_j|^2 + |S|^2 = 3\sum_{j=1}^{n} \frac{1}{|z_j|^2} + \left|\sum_{j=1}^{n} \frac{1}{z_j}\right|^2, \tag{9}$$

which is the right-hand side of (2) exactly.

Step 4: the inequality. Combine (8), (9) and Lemma 2 applied to $X = K$:

$$\sum_k \frac{1}{|w_k|^2} = \sum_k |\lambda_k(K)|^2 \leq \|K\|_F^2 = 3\sum_j \frac{1}{|z_j|^2} + \Big|\sum_j \frac{1}{z_j}\Big|^2,$$

with equality if and only if K is normal. This is (2) for every n.

Step 5: K is normal exactly when the z_j are collinear with the origin. We have $KK^* = Q\,(UM\overline{U})\,Q$ and $K^*K = Q\,(\overline{U}MU)\,Q$ with Q invertible, so K is normal if and only if $UM\overline{U} = \overline{U}MU$. Entrywise, U being diagonal and $M_{jk} = 1 + \delta_{jk}$,

$$(UM\overline{U})_{jk} = (1 + \delta_{jk})\, u_j \overline{u_k}, \qquad (\overline{U}MU)_{jk} = (1 + \delta_{jk})\, \overline{u_j} u_k.$$

The diagonal entries agree always, both being $2|u_j|^2$. Off the diagonal the factor is 1, so normality says precisely that $u_j \overline{u_k} \in \mathbb{R}$ for all $j \neq k$. Writing $u_j = \rho_j e^{i\varphi_j}$ with $\rho_j > 0$, this says $\varphi_j \equiv \varphi_k \pmod{\pi}$ for all j, k, a transitive condition; that is, all u_j lie on one line through 0. Since $1/(re^{i\theta}) = r^{-1}e^{-i\theta}$, the u_j lie on a line through 0 if and only if the z_j do. Combining with Step 4 proves Theorem 1. $\square$

Remark. The "if" half is also visible directly: if $z_j = \rho r_j$ with r_j real, then $U = \rho^{-1}R$ with R real diagonal, so $K = \rho^{-1}(QRQ)$ is a scalar times a real symmetric matrix, hence normal.

Remark (method). Steps 3 and 5 transpose a template published earlier by one of the two authors of [4] and cited there, namely [3], where the projection $S = I - \frac{1}{n}J$ plays the role of Q and the same passage from normality to collinearity is carried out; the step added here is the use of the square root $Q = M^{1/2}$, $M = I + J$ not being idempotent.

4. Two instances

Write $L(z)$ and $R(z)$ for the two sides of (2).

Equality at $n = 3$, $z = (1, 2, 3)$. Here $n + 1 = 4$ is a square, so $c = \frac{1}{3}$ exactly and everything is rational. $P(z) = z^4 - 6z^3 + 11z^2 - 6z$ and $P'(z) = 4z^3 - 18z^2 + 22z - 6$, so the elementary symmetric functions of w_1, w_2, w_3 are $(e_1, e_2, e_3) = (\frac{9}{2}, \frac{11}{2}, \frac{3}{2})$ and, all w_k being real,

$$L(z) = \sum_k \frac{1}{w_k^2} = \frac{e_2^2 - 2e_1e_3}{e_3^2} = \frac{\frac{121}{4} - \frac{27}{2}}{\frac{9}{4}} = \frac{67}{9},$$
$$R(z) = 3 \cdot \frac{49}{36} + \Big(\frac{11}{6}\Big)^2 = \frac{67}{9}.$$

Through the matrix of (5): with $u = (1, \frac{1}{2}, \frac{1}{3})$, $S = \frac{11}{6}$ and $c = \frac{1}{3}$,

$$K = \frac{1}{54}\begin{pmatrix} 101 & 38 & 35 \\ 38 & 56 & 26 \\ 35 & 26 & 41 \end{pmatrix}, \qquad \|K\|_F^2 = \frac{21708}{2916} = \frac{67}{9},$$

the squared numerators summing to 21708. This K is real symmetric, hence normal, as Step 5 requires.

Strict at $n = 2$, $z = (1, i)$. $u_1\overline{u_2} = i \notin \mathbb{R}$, so K is not normal. $P'(z) = 3z^2 - 2(1 + i)z + i$, and $L = 6 < 8 = R$.

5. Verification

The algebraic identities used above, and every number displayed in §4, are re-derived by the program `verify.py` accompanying this note, in exact arithmetic over $\mathbb{Q}(\sqrt{n+1}\,)(i)$ with no floating point anywhere and no third-party package; its recorded output is `verify.output.txt` (67 checks, all passing). Checked there are the identity $nc^2 + 2c - 1 = 0$ and $Q^2 = I + J$ symbolically for $n \le 12$; the determinant identity (6), the inverse (7), the similarity $KQ = QB$ and the Frobenius identity (9) on batteries of Gaussian-rational tuples; the equivalence "K normal $\iff$ collinear with the origin" on a battery with both classes non-empty; and an exact decision of (2) at $n = 2$ on 250 Gaussian-rational pairs, where the inequality holds always and equality occurs exactly on the collinear pairs. The program does not prove Lemma 2, and it decides (2) only at $n = 2$; for $n \geq 3$ both halves of Theorem 1 rest on the proof of §3 and not on any computation. Nothing is claimed here about the other Schoenberg type inequalities of [4], of orders -1, -4, $-2m$ and $-p$.

References

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[4] Q. Tang and T. Zhang, *Sharp Schoenberg type inequalities and the de Bruin–Sharma problem*, arXiv:2508.10341v3, 2025. arXiv:2508.10341. Conjecture C1 is as numbered in the v3 preprint.